

Sheth L.U.J & Sir M.V College of Science

Subject :- Artificial Intelligence

Practical no 3

Aim :- Decision Tree Learning

1. Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
2. Evaluate the accuracy and effectiveness of the decision tree on test data.
3. Visualize and interpret the generated decision tree.

Description :-

The program implements a Decision Tree Classification algorithm using the Breast Cancer Wisconsin dataset. The model learns patterns from the training data and predicts whether a tumor is Benign or Malignant.

Step	Code / Function	Description
1	<code>import pandas as pd</code>	Imports Pandas for reading and processing the dataset.
2	<code>import matplotlib.pyplot as plt</code>	Imports Matplotlib for displaying the decision tree visualization.
3	<code>from sklearn.model_selection import train_test_split</code>	Splits the dataset into training and testing data.
4	<code>DecisionTreeClassifier</code>	Creates the Decision Tree classification model.
5	<code>accuracy_score</code>	Calculates the accuracy of the model.
6	<code>classification_report</code>	Displays precision, recall, and F1-score.
7	<code>confusion_matrix</code>	Shows correct and incorrect predictions.
8	<code>pd.read_csv("data.csv")</code>	Loads the Breast Cancer dataset from the CSV file.

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9	<code>df.drop("id", axis=1)</code>	Removes the unnecessary ID column.
10	<code>df["diagnosis"].map()</code>	Converts M and B into numerical values: Malignant = 1 and Benign = 0.
11	<code>X = df.drop("diagnosis", axis=1)</code>	Stores the input features in X .
12	<code>y = df["diagnosis"]</code>	Stores the target/output variable in y .
13	<code>train_test_split()</code>	Divides data into 80% training and 20% testing data.
14	<code>DecisionTreeClassifier(criterion="entropy")</code>	Creates the Decision Tree using entropy as the splitting criterion.
15	<code>model.fit(X_train, y_train)</code>	Trains the Decision Tree using training data.
16	<code>model.predict(X_test)</code>	Predicts the classes for the test data.
17	<code>accuracy_score()</code>	Calculates the prediction accuracy.
18	<code>classification_report()</code>	Evaluates precision, recall, and F1-score.
19	<code>confusion_matrix()</code>	Displays the number of correct and incorrect classifications.
20	<code>plot_tree()</code>	Generates a graphical representation of the trained Decision Tree.
21	<code>plt.show()</code>	Displays the Decision Tree visualization.

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Visualization table :-

The practical produces several useful outputs:

Visualization / Output	Purpose	What it Shows
Dataset Table	Understand the dataset	Features and diagnosis values
Training/Test Split	Prepare data for ML	80% training and 20% testing
Accuracy	Measure overall performance	Percentage of correctly classified samples
Classification Report	Detailed evaluation	Precision, Recall and F1-score
Confusion Matrix	Analyze predictions	Correct and incorrect Benign/Malignant predictions
Decision Tree	Visualize the model	Features, conditions, samples and predicted classes

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Code & Output :-

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T092 Practica no 3.py - C:/Users/Vinod Mali/Desktop/AI/T092 Practica no 3.py (3.14.6)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix

df = pd.read_csv("data.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

df = df.drop("id", axis=1)

if "Unnamed: 32" in df.columns:
    df = df.drop("Unnamed: 32", axis=1)

df["diagnosis"] = df["diagnosis"].map({
    "M": 1,
    "B": 0
})

X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

print("\nNumber of Features:", X.shape[1])
print("Number of Samples:", X.shape[0])

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining Samples:", X_train.shape[0])
print("Testing Samples:", X_test.shape[0])

model = DecisionTreeClassifier(
    criterion="entropy",
    max_depth=4,
    random_state=42
)

model.fit(X_train, y_train)

print("\nDecision Tree trained successfully!")

y_pred = model.predict(X_test)

print("\nPredicted Values:")
```

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print("\nPredicted Values:")
print(y_pred)

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=["Benign", "Malignant"]
    )
)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(20, 10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=["Benign", "Malignant"],
    filled=True,
    rounded=True,
    fontsize=10
)

plt.title("Decision Tree - Breast Cancer Classification")

plt.show()
```

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OutPut :-

```
"IDLE Shell 3.14.6"
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Vinod Mali/Desktop/AI/T092 Practicla no 3.py =====
First 5 rows:
   id diagnosis  ... fractal_dimension_worst  Unnamed: 32
0  842302      M  ...             0.11890      NaN
1  842517      M  ...             0.08902      NaN
2  84300903     M  ...             0.08758      NaN
3  84348301     M  ...             0.17300      NaN
4  84358402     M  ...             0.07678      NaN

[5 rows x 33 columns]

Dataset Shape:
(569, 33)

Number of Features: 30
Number of Samples: 569

Training Samples: 455
Testing Samples: 114

Decision Tree trained successfully!

Predicted Values:
[0 1 0 1 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 0 0
 0 1 0 0 0 0 0 0 1 0 1 0 0 1 1 0 0 0 1 0 0 0 1 0 0 0 1 1 0 0 0 1 1 0
 0 0 1 0 1 1 1 0 0 0 0 1 1 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0
 0 0 1]

Accuracy: 0.9298245614035088
Accuracy Percentage: 92.98245614035088 %

Classification Report:
      precision    recall  f1-score   support

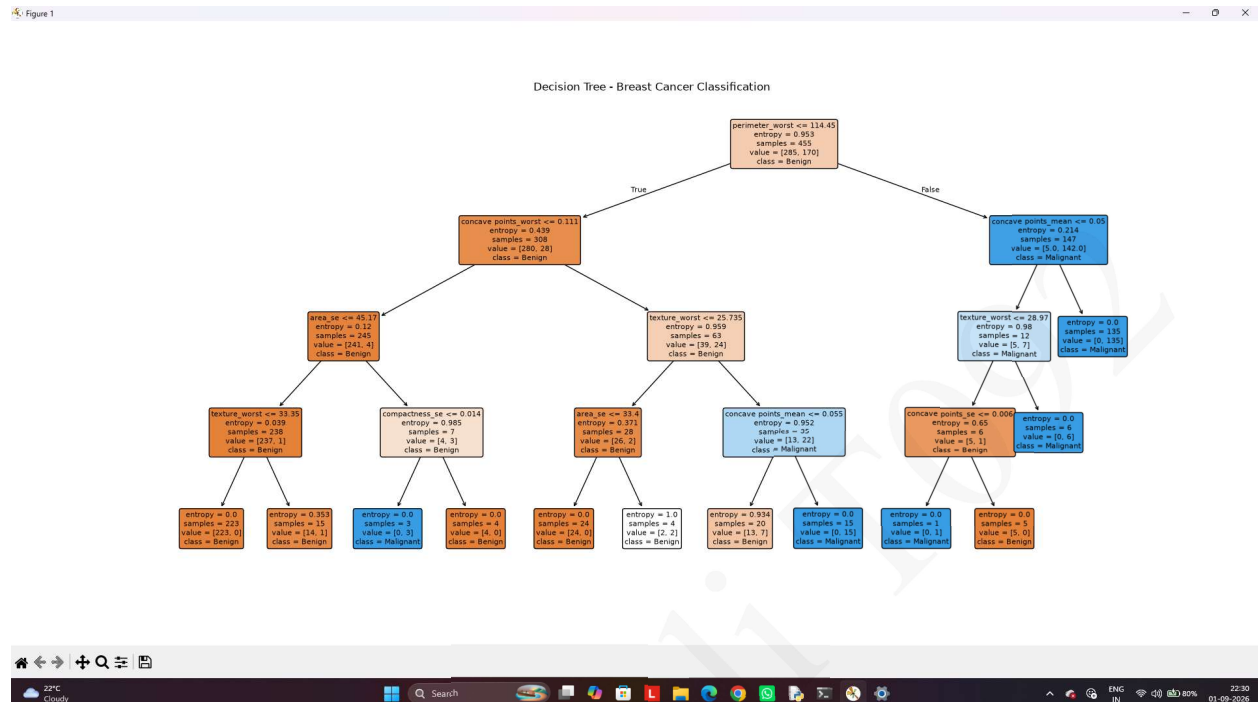
   Benign       0.90      1.00      0.95        72
  Malignant      1.00      0.81      0.89        42

   accuracy              0.93        114
  macro avg       0.95      0.90      0.92        114
 weighted avg       0.94      0.93      0.93        114

Confusion Matrix:
[[72  0]
 [ 8 34]]
```

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